# CHAPTER – 4
# IMPLEMENTATION

The project is implemented in three primary modules, each responsible for a distinct phase of the crop disease detection pipeline. The entire implementation was done in Google Colab using Python and deep learning libraries such as PyTorch, TorchVision, and the timm library for pre-trained models.

## Module 1: Data Preparation and Augmentation

This module handles the collection, structuring, and preprocessing of banana leaf images. It includes:

### 4.1.1 Data Structuring

- The dataset is organized into three main folders: Train, Validation, and Test.
- Each of these contains subfolders representing the five disease categories:
    - black_sigatoka
    - yellow_sigatoka
    - healthy
    - potassium_deficiency
    - panama_disease

```
BananaLeafDataset/
    ├── train/
    │   ├── black_sigatoka/
    │   ├── yellow_sigatoka/
    │   ├── healthy/
    │   ├── potassium_deficiency/
    │   └── panama_disease/
    ├── val/
    │   └── same structure as train
    └── test/
        └── same structure as train
```

Fig4.1.1.1 Data Structure

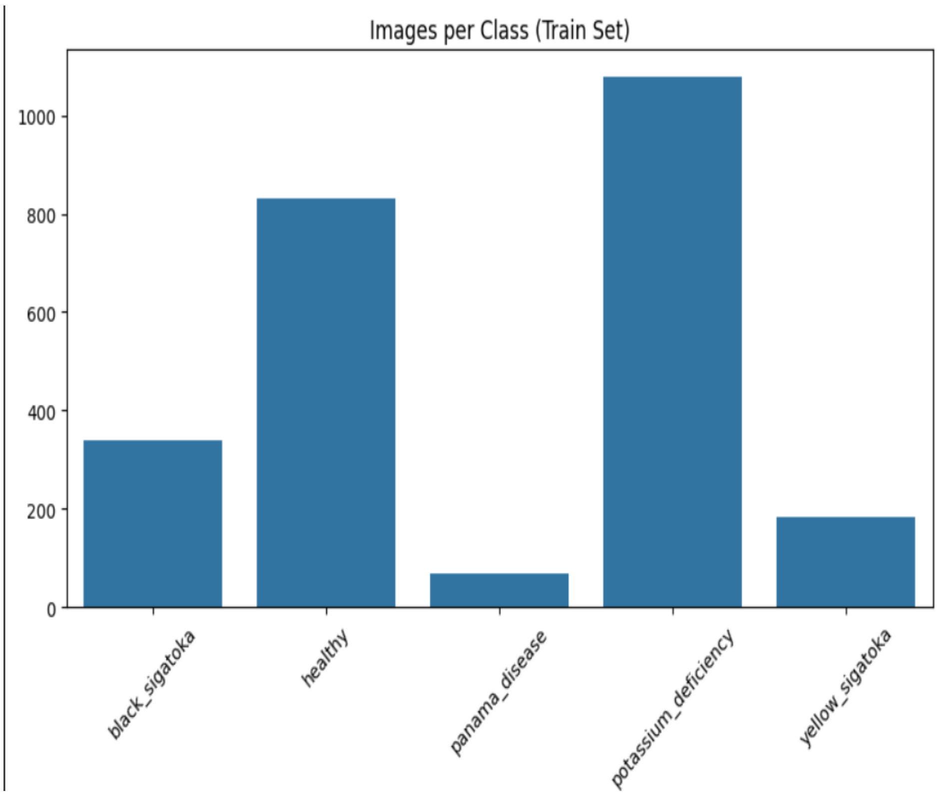


Fig. 4.1.1.2 Class Distribution in Training Set

To ensure a balanced dataset, we analyzed the number of images available for each class in the training set.

### 4.1.2 Image Preprocessing and Augmentation

- Resized all images to 224x224 pixels to match input requirements of CNN-based models.
- Applied ToTensor() transformation to convert images into normalized PyTorch tensors.
- Image augmentation (e.g., horizontal flip, random rotation) was performed to enhance dataset variability (especially useful for small datasets).

```
Found 2504 images belonging to 5 classes.
Found 573 images belonging to 5 classes.
Found 539 images belonging to 5 classes.
```

Fig 4.1.2.1Data Loading Output


Fig 4.1.2.2 Image Augmentation